# LegalGPT-India++: A Curriculum Learning and QLoRA-Based Multi-Domain Legal Question Answering Framework for Indian Constitutional, Criminal, and Procedural Law

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**Abstract**

Large Language Models (LLMs) have demonstrated remarkable success across a wide range of natural language processing tasks; however, their direct application to legal domains remains challenging due to the complexity, precision, and domain-specific nature of legal language. General-purpose language models frequently struggle to interpret statutory provisions, legal terminology, and jurisdiction-specific concepts, limiting their suitability for practical legal assistance. To address these challenges, this research presents LegalGPT-India++, a specialized legal question answering framework designed for the Indian legal ecosystem. The proposed system is built upon the Llama-3-8B foundation model and fine-tuned using the Quantized Low-Rank Adaptation (QLoRA) technique within the Unsloth framework to achieve efficient domain adaptation while minimizing computational requirements. The training corpus comprises 12,296 legal question-answer pairs collected from three major legal sources: the Constitution of India, the Indian Penal Code (IPC), and the Code of Criminal Procedure (CrPC). The final dataset includes 3,435 constitutional law samples, 2,015 criminal law samples, and 6,846 procedural law samples.

To enhance domain learning, a curriculum learning strategy was employed in which legal knowledge was introduced progressively across multiple legal domains. Additionally, a weighted dataset construction mechanism was utilized to emphasize legally significant provisions and comprehensive explanatory answers. The framework incorporates explainable prompting techniques that enable the model to generate not only legal responses but also corresponding domain and source information. Experimental evaluation demonstrates competitive performance across lexical, semantic, and domain-awareness metrics. LegalGPT-India++ achieved a BLEU score of 0.208, ROUGE-1 score of 0.531, ROUGE-2 score of 0.428, ROUGE-L score of 0.481, BERTScore F1 score of 0.895, and domain classification accuracy of 92

**Keywords:** Legal Large Language Models (Legal LLMs), LegalGPT-India++, Indian Legal AI, QLoRA Fine-Tuning, Llama-3-8B, Curriculum Learning, Legal Question Answering, Constitutional Law, Indian Penal Code, Criminal Procedure Code, Explainable Artificial Intelligence, Domain Classification, Parameter-Efficient Fine-Tuning, Legal NLP.

## 1. Introduction

The rapid evolution of artificial intelligence and natural language processing has significantly transformed the way users interact with digital information systems. Among recent technological advances, Large Language Models (LLMs) have emerged as a powerful class of models capable of understanding, generating, summarizing, and reasoning over human language. Transformer-based architectures such as GPT, LLaMA, and related foundation models have demonstrated impressive performance across diverse applications including text generation, question answering, information retrieval, and conversational systems. Despite these achievements, the deployment of such models within domain-specific environments remains a considerable challenge, particularly in the legal sector where accuracy, interpretability, and contextual understanding are essential requirements.

The legal domain differs substantially from general-purpose language environments due to the presence of specialized terminology, complex statutory structures, formal drafting conventions, and jurisdiction-specific knowledge. Legal questions often require precise interpretation of legislative provisions and procedural rules, making it difficult for generic language models to consistently generate reliable responses. These limitations become even more pronounced within the Indian legal ecosystem, which consists of a vast collection of constitutional provisions, criminal statutes, procedural regulations, judicial doctrines, and administrative laws.

In India, legal information is frequently accessed by legal professionals, policymakers, students, researchers, and ordinary citizens seeking guidance regarding rights, obligations, and legal procedures. However, navigating lengthy constitutional articles and statutory provisions is often time-consuming and requires substantial legal expertise. Consequently, there is a growing demand for intelligent legal systems capable of providing accurate, accessible, and explainable responses to legal queries. Existing legal artificial intelligence systems predominantly focus on Western jurisdictions and therefore exhibit limited suitability for the unique legal structure of India.

To address this research gap, this study proposes LegalGPT-India++, a domain-specific legal question answering framework tailored for Indian law. The proposed system utilizes the Llama-3-8B foundation model and adopts Quantized Low-Rank Adaptation (QLoRA) for parameter-efficient fine-tuning. A curriculum learning strategy is employed to progressively expose the model to Constitutional Law, Criminal Law, and Criminal Procedure Law concepts. The training corpus consists of 12,296 legal question-answer pairs derived from the Constitution of India, the Indian Penal Code, and the Code of Criminal Procedure. Furthermore, a weighted dataset engineering approach is implemented to prioritize legally significant provisions and improve knowledge retention.

The effectiveness of the proposed framework is evaluated using lexical overlap metrics, semantic similarity measures, and legal domain classification performance. Experimental results demonstrate strong semantic understanding with a BERTScore F1 of 0.895 and a domain classification accuracy of 92

## 2. Related Work

The application of artificial intelligence within the legal domain has evolved considerably over the past decade. Earlier legal information systems primarily relied on rule-based approaches, keyword matching, ontology-driven frameworks, and legal knowledge bases. While these systems were capable of retrieving legal information, they often struggled to understand contextual relationships and generate human-like responses. Recent advances in machine learning and deep learning have significantly improved the effectiveness of legal information retrieval and legal question answering systems.

### 2.1 Legal Question Answering Systems

Legal Question Answering (Legal QA) systems aim to provide accurate responses to legal queries by retrieving, interpreting, and generating legally relevant information. Traditional Legal QA systems largely depended on information retrieval techniques and expert-defined legal rules. Although effective for structured legal databases, these approaches suffered from limited scalability and poor contextual understanding.

Recent research has introduced neural architectures capable of learning complex language patterns from legal corpora. These systems improve response quality by leveraging pre-trained language models and domain-specific training data. However, many existing Legal QA systems remain focused on legal document retrieval rather than explainable response generation.

### 2.2 Legal Large Language Models

The emergence of Large Language Models (LLMs) has transformed legal natural language processing. Models such as GPT, BERT, RoBERTa, and LLaMA have demonstrated strong performance in text generation, summarization, classification, and question answering.

Several legal-domain adaptations, including Legal-BERT and other domain-specific transformer models, have shown that specialized legal pre-training substantially improves legal language understanding. Nevertheless, most existing legal LLMs have been developed using legal data from Western jurisdictions, resulting in limited applicability to Indian legal frameworks.

## 2.3 Transformer-Based Legal AI

Transformer architectures have become the dominant paradigm for legal artificial intelligence applications. Self-attention mechanisms allow transformer models to capture long-range dependencies frequently present within legal texts, statutes, and regulations.

In legal environments, transformer models have been successfully applied to legal judgment prediction, legal document classification, statute retrieval, contract analysis, and legal summarization. Their ability to model contextual information makes them particularly suitable for legal reasoning tasks involving complex statutory provisions.

## 2.4 Parameter-Efficient Fine-Tuning

Although foundation models demonstrate impressive capabilities, full fine-tuning often requires substantial computational resources. Parameter-Efficient Fine-Tuning (PEFT) techniques such as LoRA, QLoRA, Adapter Tuning, and Prefix Tuning have emerged as practical alternatives.

Among these methods, QLoRA enables memory-efficient training by employing low-rank adapters and quantized model representations. This approach significantly reduces hardware requirements while preserving competitive model performance, making it highly attractive for domain-specific legal model development.

## 2.5 Research Gap Analysis

Current legal language models exhibit several limitations. First, most models focus primarily on legal systems outside India. Second, many legal question answering systems lack explainability and source attribution mechanisms. Third, few studies combine curriculum learning, weighted dataset engineering, domain classification, and parameter-efficient fine-tuning within a unified framework.

To address these limitations, LegalGPT-India++ proposes a curriculum-learning-driven, QLoRA-based legal language model specifically designed for Indian Constitutional Law, Indian Penal Code, and Code of Criminal Procedure domains while incorporating explainable legal response generation and domain-aware reasoning.

# 3. Dataset Construction and Preprocessing

The performance of domain-specific language models is highly dependent upon the quality, diversity, and representativeness of the training data. To develop an effective legal question answering framework, a multi-domain Indian legal corpus was constructed using authoritative legal sources.

## 3.1 Dataset Overview

The final training corpus contains 12,296 question-answer pairs spanning multiple legal domains. The dataset was specifically designed to support legal question answering, domain classification, and explainable response generation.

Each sample contains five primary fields:

- Question
- Answer
- Domain
- Source
- Sample Weight

The structured format enables both supervised learning and legal source attribution.

## 3.2 Constitution Dataset

The Constitutional Law subset contains 3,435 question-answer pairs derived from the Constitution of India. These samples cover topics including fundamental rights, fundamental duties, citizenship, constitutional remedies, legislative powers, executive functions, judiciary provisions, and federal governance structures.

The constitutional corpus provides foundational legal knowledge necessary for understanding higher-level legal concepts.

## 3.3 IPC Dataset

The Indian Penal Code (IPC) component contains 2,015 samples focusing on criminal offences, punishments, criminal liability, legal definitions, and statutory criminal provisions.

The IPC dataset enables the model to learn offence classification, punishment interpretation, and criminal law concepts frequently encountered in legal practice.

## 3.4 CrPC Dataset

The Code of Criminal Procedure (CrPC) subset represents the largest portion of the corpus with 6,846 training samples.

The dataset focuses on procedural legal matters including:

- Arrest procedures
- Investigation processes
- Bail provisions

- Magistrate powers
- Trial procedures
- Procedural safeguards

This extensive procedural coverage supports practical legal question answering scenarios.

### 3.5 Dataset Statistics

The complete dataset distribution is summarized below.

Table 1: Dataset Distribution

| Domain | Samples |
|---|---|
| Constitution | 3435 |
| IPC | 2015 |
| CrPC | 6846 |
| Total | 12296 |

### 3.6 Data Cleaning

Several preprocessing operations were performed before model training:

- Duplicate removal
- Text normalization
- Invalid record filtering
- Formatting correction
- Missing-value handling

These operations improved dataset consistency while preserving legally significant terminology.

### 3.7 Data Augmentation

Question paraphrasing and answer reformulation techniques were utilized where appropriate to improve linguistic diversity. This process enabled the model to learn multiple formulations of similar legal concepts without altering legal meaning.

## 3.8 Weighted Dataset Generation

A weighted dataset strategy was implemented to emphasize important legal content. Higher weights were assigned to legally significant provisions, longer explanatory answers, and conceptually important constitutional articles.

This weighting mechanism enabled the model to focus more strongly on critical legal concepts during optimization.

## 3.9 Curriculum Dataset Preparation

To support curriculum learning, training samples were organized according to increasing legal complexity.

The learning sequence consisted of:

1. Constitutional Law
2. Indian Penal Code
3. Code of Criminal Procedure

This staged progression allowed gradual acquisition of foundational legal knowledge before introducing criminal and procedural legal reasoning tasks.

# 4. Proposed Methodology

LegalGPT-India++ is designed as a domain-adapted legal language model capable of answering Indian legal questions while simultaneously identifying legal domains and legal sources. The framework integrates curriculum learning, QLoRA-based fine-tuning, explainable prompting, and domain-aware classification within a unified architecture.

## 4.1 LegalGPT-India++ Architecture

The proposed architecture consists of four primary components:

1. Foundation Language Model
2. Curriculum Learning Module
3. QLoRA Adaptation Layer
4. Explainable Response Generator

Together, these components enable efficient legal knowledge acquisition and transparent response generation.

## 4.2 Llama-3-8B Foundation Model

Llama-3-8B serves as the foundation model for LegalGPT-India++. The model was selected due to its strong language understanding capabilities, scalability, and suitability for domain adaptation tasks.

Its transformer-based architecture provides the contextual understanding required for legal question answering.

## 4.3 QLoRA Fine-Tuning

Rather than performing full fine-tuning, Quantized Low-Rank Adaptation (QLoRA) was employed.

Advantages include:

- Reduced memory requirements
- Faster training
- Lower computational cost
- Competitive downstream performance

QLoRA enables efficient adaptation of large language models using lightweight trainable parameters.

## 4.4 Unsloth Optimization Framework

The Unsloth framework was utilized to accelerate training and improve memory efficiency.

Benefits of Unsloth include:

- Faster fine-tuning
- Reduced GPU memory consumption
- Efficient large-model adaptation

These optimizations were particularly important for experimental feasibility.

## 4.5 Curriculum Learning Framework

Curriculum learning was implemented to introduce legal concepts gradually.

The learning progression was structured as:

1. Constitutional Law
2. Criminal Law (IPC)
3. Criminal Procedure Law (CrPC)

This ordering allows the model to build foundational legal understanding before learning more complex procedural concepts.

### 4.6 Prompt Engineering Design

Training prompts were designed to enforce structured legal outputs.

Each training sample followed a standardized format:

```
Question:
Domain:
Source:
Answer:
```

This prompt template encourages both legal reasoning and explainability.

### 4.7 Explainable Output Structure

A defining feature of LegalGPT-India++ is its explainable response format.

Each generated answer contains:

- Legal Domain
- Legal Source
- Final Answer

This improves transparency and user confidence in generated legal information.

### 4.8 Domain Classification Strategy

In addition to answer generation, the model learns domain classification.

The three target domains include:

1. Constitutional Law
2. Indian Penal Code (IPC)
3. Code of Criminal Procedure (CrPC)

## 5. Experimental Setup

This section describes the computational environment, training configuration, optimization strategy, and evaluation procedures used to develop and assess the LegalGPT-India++ framework. The experimental design was structured to evaluate the effectiveness of curriculum learning and QLoRA-based fine-tuning for Indian legal question answering while maintaining computational efficiency.

## 5.1 Hardware Configuration

All experiments were conducted using GPU-enabled computational resources suitable for large language model fine-tuning. The use of QLoRA significantly reduced the memory requirements associated with traditional full-parameter fine-tuning.

The hardware infrastructure was selected to ensure:

- Stable large-model training
- Efficient memory utilization
- Reduced computational cost
- Support for parameter-efficient adaptation

The adoption of quantization and low-rank adaptation enabled the Llama-3-8B model to be fine-tuned within resource-constrained research environments.

## 5.2 Software Environment

The software stack consisted primarily of modern transformer-based deep learning frameworks and optimization libraries.

The experimental environment included:

- Python
- PyTorch
- Hugging Face Transformers
- Unsloth Framework
- PEFT Library
- QLoRA Adapters
- CUDA-enabled GPU Runtime

These tools collectively enabled scalable instruction tuning and efficient legal-domain adaptation.

## 5.3 Training Configuration

The training corpus contained 12,296 legal question-answer pairs spanning Constitutional Law, Indian Penal Code (IPC), and Code of Criminal Procedure (CrPC) domains.

The training process was designed around:

1. Dataset preprocessing

2. Weighted sample generation
3. Curriculum-based ordering
4. Instruction fine-tuning
5. Evaluation

Training samples were formatted using a standardized prompt structure to encourage domain-aware and explainable legal response generation.

## 5.4 Hyperparameters

The major training hyperparameters are summarized below.

Table 2: Training Hyperparameters

| Parameter | Value |
|---|---|
| Base Model | Llama-3-8B |
| Fine-Tuning Method | QLoRA |
| Framework | Unsloth |
| Training Epochs | 1 |
| Dataset Size | 12,296 |
| Final Training Loss | 1.456 |

Training history demonstrated a consistent reduction in loss values, indicating stable optimization and successful legal-domain adaptation.

## 5.5 Training Pipeline

The complete training workflow followed the sequence:

1. Data Collection
2. Data Cleaning
3. Sample Weight Assignment
4. Curriculum Arrangement
5. Prompt Formatting
6. QLoRA Fine-Tuning
7. Evaluation

This pipeline enabled progressive acquisition of constitutional, criminal, and procedural legal knowledge.

## 5.6 Evaluation Protocol

Model evaluation was performed on held-out legal samples using lexical overlap metrics, semantic similarity measures, and legal domain classification accuracy.

The evaluation considered three complementary perspectives:

- Lexical correctness
- Semantic correctness
- Legal domain awareness

This multi-metric evaluation strategy provided a comprehensive assessment of model performance.

# 6. Evaluation Metrics

To comprehensively evaluate LegalGPT-India++, both lexical and semantic evaluation measures were employed. Since legal responses may exhibit different wording while preserving identical legal meaning, multiple complementary metrics were utilized to assess performance.

## 6.1 BLEU

The Bilingual Evaluation Understudy (BLEU) metric measures the extent of n-gram overlap between generated responses and ground-truth legal answers.

BLEU is calculated as:

$$BLEU = BP \times \exp\left(\sum_{n=1}^{N} w_n \log p_n\right) \tag{1}$$

where:

- $BP$ represents brevity penalty
- $p_n$ represents modified n-gram precision
- $w_n$ denotes the weight assigned to each n-gram level

BLEU evaluates lexical similarity between generated and reference responses.

## 6.2 ROUGE-1

ROUGE-1 measures unigram overlap between reference and generated texts.

The metric captures how effectively important legal keywords and concepts are recovered within generated responses.

$$ROUGE\text{-}1 = \frac{\text{Overlapping Unigrams}}{\text{Reference Unigrams}} \tag{2}$$

## 6.3 ROUGE-2

ROUGE-2 evaluates bigram overlap and captures phrase-level similarity.

$$ROUGE\text{-}2 = \frac{\text{Overlapping Bigrams}}{\text{Reference Bigrams}} \tag{3}$$

This measure is particularly useful in legal text evaluation because many legal concepts depend upon contextual phrase structures.

## 6.4 ROUGE-L

ROUGE-L utilizes the Longest Common Subsequence (LCS) between generated and reference answers.

$$ROUGE\text{-}L = \frac{LCS(X,Y)}{|Y|} \tag{4}$$

where:

- $LCS(X, Y)$ denotes the longest common subsequence
- $Y$ represents the reference answer

This metric evaluates structural consistency in generated legal responses.

## 6.5 BERTScore Precision

BERTScore Precision measures semantic relevance by comparing contextual embeddings generated by transformer models.

$$Precision = \frac{\sum_{x\in C} \max_{y\in R} x^T y}{|C|} \tag{5}$$

where:

- $C$ denotes candidate tokens
- $R$ denotes reference tokens

## 6.6 BERTScore Recall

BERTScore Recall evaluates how much semantic information from the reference answer is successfully captured.

$$Recall = \frac{\sum_{y\in R} \max_{x\in C} x^T y}{|R|} \tag{6}$$

## 6.7 BERTScore F1

The BERTScore F1 metric combines precision and recall to provide an overall semantic similarity measure.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{7}$$

BERTScore is particularly valuable in legal NLP because semantically equivalent answers may use different wording.

## 6.8 Domain Classification Accuracy

To evaluate legal-domain awareness, classification accuracy was calculated.

$$Accuracy = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100 \tag{8}$$

The model classified questions into:

1. Constitutional Law
2. Indian Penal Code (IPC)
3. Code of Criminal Procedure (CrPC)

This metric evaluates the effectiveness of domain-aware legal reasoning.

# 7. Results and Discussion

This section presents the experimental findings and discusses the effectiveness of LegalGPT-India++ across multiple evaluation dimensions. The obtained results demonstrate the capability of the framework to learn meaningful legal knowledge representations while maintaining strong semantic consistency with authoritative legal references.

## 7.1 Quantitative Results

The overall model performance is summarized in Table 3.

Table 3: Performance Results

| Metric | Score |
|---|---|
| BLEU | 0.208 |
| ROUGE-1 | 0.531 |
| ROUGE-2 | 0.428 |
| ROUGE-L | 0.481 |
| BERT Precision | 0.869 |
| BERT Recall | 0.923 |
| BERT F1 | 0.895 |
| Domain Accuracy | 92% |

The results indicate strong semantic performance and effective legal-domain adaptation.

### 7.2 Training Convergence Analysis

Training loss exhibited a gradual reduction throughout the optimization process.

The initial loss value was approximately:

$$Loss_{initial} \approx 1.60 \tag{9}$$

The final recorded loss was:

$$Loss_{final} \approx 1.456 \tag{10}$$

This reduction demonstrates successful knowledge acquisition and stable optimization behavior during QLoRA fine-tuning.

### 7.3 Semantic Performance Analysis

The strongest performance was observed in semantic evaluation metrics.

The BERTScore F1 value of:

$$BERTScore_{F1} = 0.895 \tag{11}$$

demonstrates that generated responses preserve legal meaning even when wording differs from reference answers.

This result is particularly important in legal question answering because multiple legally correct responses may exist for a given query.

### 7.4 Domain Classification Results

The model achieved:

$$Accuracy = 92\% \tag{12}$$

for legal-domain prediction.

This indicates effective discrimination among:

- Constitutional Law
- IPC
- CrPC

The result validates the effectiveness of the domain-aware learning strategy introduced in LegalGPT-India++.

## 7.5 Explainability Analysis

A major contribution of the proposed framework is explainability.

Unlike traditional legal QA systems that generate only answers, LegalGPT-India++ produces:

1. Domain Identification
2. Source Attribution
3. Final Legal Answer

This structure improves transparency and supports user verification of generated responses.

## 7.6 Comparative Discussion

Lexical metrics such as BLEU and ROUGE indicate moderate textual overlap, whereas semantic metrics demonstrate substantially higher performance.

This observation suggests that LegalGPT-India++ prioritizes semantic correctness rather than merely reproducing exact wording.

Such behavior is desirable in legal applications where preserving legal meaning is often more important than reproducing identical statutory text.

## 7.7 Error Analysis

Although performance is encouraging, several weaknesses remain.

Observed errors include:

- Repetitive response generation
- Incomplete statutory references

- Domain confusion in rare cases
- Over-representation of CrPC concepts

The dominance of procedural-law samples within the training corpus may contribute to these issues.

Future improvements may include:

1. Dataset balancing
2. Multi-epoch training
3. Retrieval-Augmented Generation (RAG)
4. Judicial precedent integration

## 8. Case Studies

While automatic evaluation metrics provide quantitative insights into model performance, qualitative analysis is equally important for understanding the practical effectiveness of legal question answering systems. This section presents representative case studies demonstrating the capabilities and limitations of LegalGPT-India++ across multiple legal domains.

### 8.1 Constitutional Law Queries

The model demonstrated strong performance when answering questions related to Constitutional Law. Queries involving Fundamental Rights, Directive Principles of State Policy, Fundamental Duties, constitutional remedies, and the powers of Parliament were generally answered correctly.

The model was able to identify relevant constitutional concepts and generate structured responses containing domain information, legal source attribution, and explanatory answers. This behavior suggests that curriculum learning effectively enabled the model to acquire foundational constitutional knowledge before exposure to more specialized legal domains.

Representative constitutional topics successfully addressed include:

- Fundamental Rights
- Article-based Constitutional Provisions
- Citizenship
- Constitutional Remedies
- Parliamentary Powers
- Federal Governance Structure

## 8.2 IPC Queries

Indian Penal Code (IPC) questions focused on criminal offences, punishments, criminal liability, and legal definitions.

For most criminal law queries, the model successfully identified:

- Nature of the offence
- Relevant legal provisions
- Punishment-related information
- Criminal law terminology

The outputs demonstrated strong semantic alignment with reference answers despite variations in wording. This indicates effective adaptation to criminal law concepts using parameter-efficient fine-tuning.

## 8.3 CrPC Queries

Since CrPC samples formed the largest component of the training dataset, the model demonstrated particularly strong performance on procedural legal questions.

Commonly successful question categories included:

- Arrest Procedures
- Bail Provisions
- Investigation Processes
- Trial Procedures
- Magistrate Powers
- Procedural Safeguards

The model showed a strong understanding of legal procedures and procedural terminology, reflecting the substantial representation of CrPC examples in the training corpus.

## 8.4 Correct Predictions

Analysis of correctly answered legal questions revealed several recurring strengths:

1. Accurate domain identification.
2. Correct legal source attribution.
3. High semantic consistency.

4. Well-structured explanations.
5. Context-aware legal reasoning.

These observations are consistent with the model's high BERTScore F1 and strong domain classification performance.

## 8.5 Failure Examples

Despite strong overall performance, certain weaknesses were observed.

Typical failure cases included:

- Confusion between closely related legal provisions.
- Partial omission of important statutory details.
- Repetitive answer generation.
- Incomplete source attribution.
- Over-generalization in complex legal scenarios.

These errors were relatively infrequent but indicate opportunities for future enhancement through retrieval-based legal grounding mechanisms and larger domain-balanced datasets.

# 9. Advantages of the Proposed System

The proposed LegalGPT-India++ framework introduces several technical and practical advantages over conventional legal question answering systems. The combination of curriculum learning, weighted dataset engineering, QLoRA adaptation, and explainable prompting contributes to improved legal-domain performance and operational efficiency.

## 9.1 Explainability

One of the most significant advantages of LegalGPT-India++ is its explainable response generation mechanism.

Unlike traditional language models that generate standalone answers, the proposed framework produces:

1. Legal Domain
2. Legal Source
3. Generated Legal Answer

This structure enables users to understand the origin of generated responses and improves transparency within legal decision-support systems.

Furthermore, explainable outputs facilitate legal verification and increase user trust in AI-assisted legal applications.

## 9.2 Computational Efficiency

Large-scale language model fine-tuning often requires substantial computational resources. The use of QLoRA and the Unsloth framework significantly reduces memory requirements without sacrificing performance.

Major efficiency benefits include:

- Reduced GPU memory usage.
- Faster fine-tuning.
- Lower computational costs.
- Practical deployment on limited hardware.

These advantages make LegalGPT-India++ accessible to academic researchers and organizations with limited infrastructure.

## 9.3 Domain Awareness

A distinguishing feature of the proposed framework is its ability to identify legal domains alongside answer generation.

The system classifies legal queries into:

1. Constitutional Law
2. Indian Penal Code (IPC)
3. Code of Criminal Procedure (CrPC)

This domain-awareness improves response accuracy and enables more structured legal question answering.

The achieved domain classification accuracy of 92% demonstrates the effectiveness of this design.

## 9.4 Scalability

The architecture has been designed with scalability in mind.

Additional legal domains can be incorporated through further fine-tuning, including:

- Commercial Law
- Cyber Law
- Taxation Law
- Intellectual Property Law

- Environmental Law
- Labour Law

The curriculum-learning framework provides a systematic mechanism for extending legal knowledge coverage without redesigning the underlying architecture.

## 9.5 Practical Applicability

The proposed framework can support a wide range of legal stakeholders.

Potential application areas include:

- Legal Education
- Academic Research
- Legal Information Retrieval
- Citizen Legal Assistance
- Institutional Knowledge Support
- Policy Research

Its combination of explainability, semantic accuracy, and computational efficiency makes it a practical foundation for next-generation legal AI systems.

# 10. Limitations

Although the experimental results demonstrate promising performance, several limitations remain that should be considered when evaluating the capabilities of LegalGPT-India++.

## 10.1 Dataset Imbalance

The training corpus exhibits noticeable distribution imbalance across legal domains.

The dataset consists of:

- Constitutional Law: 3,435 samples
- IPC: 2,015 samples
- CrPC: 6,846 samples

Since CrPC contributes more than half of the total training data, the model may exhibit a bias toward procedural legal concepts.

Future studies should consider balanced sampling strategies and expanded datasets to mitigate this issue.

## 10.2 Limited Training Epochs

The current model was trained for only one epoch.

Although stable convergence was observed and a final training loss of approximately 1.456 was achieved, additional epochs may further improve knowledge retention, lexical accuracy, and response consistency.

Future work should investigate:

- Multi-epoch training
- Hyperparameter optimization
- Extended convergence analysis

to determine optimal training configurations.

## 10.3 Hallucination Risk

Like most modern Large Language Models, LegalGPT-India++ is not immune to hallucination.

Potential challenges include:

- Generation of unsupported legal statements.
- Incomplete statutory references.
- Incorrect procedural interpretations.
- Ambiguous legal explanations.

Therefore, generated responses should be viewed as informational assistance rather than authoritative legal advice.

Legal professionals should verify outputs against official legal sources before practical use.

## 10.4 Legal Scope Constraints

The proposed framework focuses exclusively on three legal domains:

1. Constitutional Law
2. Indian Penal Code
3. Code of Criminal Procedure

Consequently, the model lacks knowledge regarding several important legal disciplines such as:

- Civil Law

- Corporate Law
- Labour Law
- Taxation Law
- Cyber Law
- Intellectual Property Law

Expanding domain coverage remains an important research direction.

## 10.5 Absence of Retrieval-Augmented Generation

The current architecture relies entirely on knowledge acquired during fine-tuning.

Unlike Retrieval-Augmented Generation (RAG) frameworks, the model does not access external legal databases during inference.

As a result:

- Updates to legal provisions cannot be reflected automatically.
- Recent legislative amendments remain unavailable.
- Judicial precedents are not dynamically retrieved.

## 11. Future Work

Although LegalGPT-India++ demonstrates promising performance in legal question answering and domain-aware response generation, several research opportunities exist for extending the framework. Future improvements aim to enhance factual reliability, broaden legal coverage, improve reasoning capabilities, and support real-world deployment within legal environments.

## 11.1 Retrieval-Augmented Generation

One of the most promising directions for future work is the integration of Retrieval-Augmented Generation (RAG) techniques.

The current version of LegalGPT-India++ relies solely on knowledge learned during fine-tuning. Consequently, the model cannot dynamically access updated legal information during inference.

Future implementations could incorporate:

- Legal statutes databases
- Government legal repositories
- Parliamentary records

- Legal encyclopedias
- Official legal notifications

The integration of retrieval mechanisms would improve factual accuracy, reduce hallucinations, and ensure that answers remain consistent with the latest legal developments.

## 11.2 Judicial Precedent Integration

The present framework primarily utilizes statutory legal information.

However, judicial precedents play a critical role within the Indian legal system. Future versions may include:

- Supreme Court Judgments
- High Court Judgments
- Landmark Constitutional Cases
- Criminal Law Precedents
- Legal Commentaries

Incorporating case law would improve legal reasoning and enable the model to generate responses supported by judicial interpretations rather than statutory provisions alone.

## 11.3 Multilingual Indian Legal AI

India is linguistically diverse, creating a need for legal assistance systems that operate across multiple languages.

Future work can focus on supporting:

- Hindi
- Bengali
- Marathi
- Tamil
- Telugu
- Gujarati
- Kannada
- Malayalam
- Punjabi

A multilingual framework would increase accessibility and enable wider adoption among legal practitioners and citizens across different regions of India.

## 11.4 Advanced Legal Reasoning

While the proposed system performs legal question answering effectively, future systems should focus on advanced reasoning capabilities.

Potential enhancements include:

- Multi-hop Legal Reasoning
- Legal Argument Generation
- Case Outcome Prediction
- Contradiction Detection
- Legal Evidence Analysis

Such improvements would enable LegalGPT-India++ to move beyond information retrieval and toward intelligent legal decision-support.

## 11.5 Expansion of Legal Domains

The current system focuses on Constitutional Law, IPC, and CrPC.

Future datasets may incorporate:

- Civil Law
- Commercial Law
- Corporate Law
- Cyber Law
- Taxation Law
- Labour Law
- Environmental Law
- Intellectual Property Law

This expansion would improve the comprehensiveness and practical utility of the framework.

## 11.6 Agentic Legal Systems

Recent advances in generative AI have enabled the emergence of autonomous AI agents.

Future research may investigate:

- Automated Legal Research Assistants
- Legal Drafting Assistants
- Case Summarization Systems
- Legal Compliance Agents
- Intelligent Judicial Support Systems

These developments could significantly enhance productivity within legal practice and legal education.

# 12. Conclusion

This research presented **LegalGPT-India++**, a domain-specific legal question answering framework designed for the Indian legal ecosystem. The proposed approach combines curriculum learning, weighted dataset engineering, explainable prompting, and parameter-efficient fine-tuning to improve legal knowledge acquisition while maintaining computational efficiency.

The framework was built upon the Llama-3-8B foundation model and adapted using Quantized Low-Rank Adaptation (QLoRA) within the Unsloth optimization environment. A curated corpus consisting of 12,296 legal question-answer pairs was constructed from three major legal domains, namely the Constitution of India, the Indian Penal Code (IPC), and the Code of Criminal Procedure (CrPC). Furthermore, a weighted dataset strategy was implemented to prioritize legally significant provisions, while a curriculum-learning methodology enabled gradual acquisition of increasingly complex legal concepts.

Experimental evaluation demonstrated that the proposed framework effectively learns legal-domain knowledge and produces semantically accurate responses. The model achieved strong performance across multiple evaluation metrics, including BLEU, ROUGE, BERTScore, and legal-domain classification accuracy. In particular, the high BERTScore F1 value indicates strong semantic alignment between generated answers and reference legal responses, while the domain classification results demonstrate successful identification of legal categories.

A major contribution of this work is the introduction of an explainable legal response generation framework. Instead of generating standalone answers, LegalGPT-India++ provides legal domain identification, source attribution, and answer generation in a structured format. This improves transparency, interpretability, and trustworthiness, which are essential requirements in legal artificial intelligence applications.

Although several limitations remain, including dataset imbalance, restricted legal coverage, and the absence of retrieval-based grounding mechanisms, the overall results validate the effectiveness of combining curriculum learning and QLoRA-based adaptation for legal language modeling.

In conclusion, LegalGPT-India++ successfully bridges the gap between general-purpose large language models and specialized Indian legal knowledge systems. The framework provides a scalable and extensible foundation for future research involving retrieval-augmented legal intelligence, judicial precedent analysis, multilingual legal assistance, advanced legal reasoning, and next-generation AI-powered legal support systems.

## A. Sample Prompt Template

A representative training prompt used during fine-tuning is shown below.

```
Question:
What is the punishment for theft under IPC?

Domain:
Indian Penal Code

Source:
Indian Penal Code

Answer:
The punishment for theft is imprisonment,
fine, or both depending upon the provisions
specified under the relevant section.
```

## B. Dataset Statistics

Table 4: Dataset Composition

| Dataset Category | Samples |
|---|---|
| Constitutional Law | 3435 |
| Indian Penal Code | 2015 |
| Code of Criminal Procedure | 6846 |
| Total Samples | 12296 |

## C. Evaluation Summary

Table 5: Overall Performance Metrics

| Metric | Value |
|---|---|
| BLEU | 0.208 |
| ROUGE-1 | 0.531 |
| ROUGE-2 | 0.428 |
| ROUGE-L | 0.481 |
| BERT Precision | 0.869 |
| BERT Recall | 0.923 |
| BERT F1 | 0.895 |
| Domain Accuracy | 92% |

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